Lecture 11 — Least Squares

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Wednesday, October 14, 2026

After this lecture you should be able to

- Set up a data-fitting problem as an overdetermined system $\boldsymbol{X}\boldsymbol{k} = \boldsymbol{y}$ and write the total squared error as $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e}$.
- Interpret least squares geometrically: find the point of $\mathrm{col}(\boldsymbol{X})$ closest to $\boldsymbol{y}$, so that the residual is orthogonal to $\mathrm{col}(\boldsymbol{X})$.
- Turn that right angle into the normal equations $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\,\overline{\boldsymbol{k}} = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$, and solve them.
- Extend the same matrix set-up to several independent variables and to polynomial models.

1 A problem you have already met

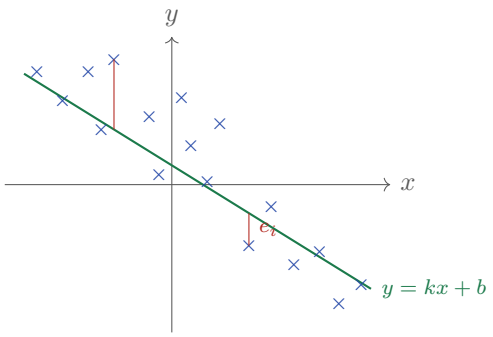
Suppose we are handed data

$$\{(x_i, y_i)\}_{i\in\{1,\dots,m\}}, \qquad x_i, y_i \in \mathbb{R}.$$

For example, x_i is a person's height and y_i is their shoe size; or x_i is the horsepower of a vehicle and y_i is its cost; etc., etc. We want to *fit a linear model*

$$y = kx + b$$

to that data.



How do we determine the parameters k and b of the line? No line passes through all the crosses — with m data points and two unknowns the system is *overdetermined* — so "solve $\boldsymbol{X}\boldsymbol{k} = \boldsymbol{y}$" is the wrong question. We must decide what "best" means first.

2 The classic criterion: minimise the total squared error

For a given data point (x_i, y_i), the prediction of our model at x_i is $kx_i + b$. The error between the model prediction and the actual observation y_i is

$$y_i - (kx_i + b),$$

and the **total squared error** is

$$\sum_{i=1}^{m} \big[\, y_i - (kx_i + b) \,\big]^2.$$

Let us write this using vectors and matrices. Collect the data into

$$\boldsymbol{X} = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \qquad \boldsymbol{k} = \begin{bmatrix} k \\ b \end{bmatrix}.$$

(The board writes these X, Y, K; the lower case here only marks which are vectors.) The column of ones is what carries the intercept b. The error as a vector is then

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{k}, \qquad e_i = y_i - (kx_i + b),$$

and the total squared error is $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e} = \sum_i e_i^2$:

$$\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e} = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{k})^{\mathsf{T}}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{k}) = \boldsymbol{y}^{\mathsf{T}}\boldsymbol{y} + \boldsymbol{k}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{k} - \boldsymbol{y}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{k} - \boldsymbol{k}^{\mathsf{T}}\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}.$$

The last two terms are 1×1, hence equal to their own transposes, so they combine:

$$\boxed{\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e} = \boldsymbol{k}^{\mathsf{T}}(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\boldsymbol{k} - 2\,\boldsymbol{y}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{k} + \boldsymbol{y}^{\mathsf{T}}\boldsymbol{y}}$$

So the least squares fitting problem can be written as the optimisation

$$\min_{\boldsymbol{k}\in\mathbb{R}^2} \; \boldsymbol{k}^{\mathsf{T}}(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\boldsymbol{k} - 2\,\boldsymbol{y}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{k} + \boldsymbol{y}^{\mathsf{T}}\boldsymbol{y}.$$

Key idea

Everything about the data has been compressed into the two objects $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}$ (a 2×2 matrix) and $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$ (a 2-vector). The number of data points m has left the problem. That collapse from m to 2 is the whole reason least squares is cheap.

Example 2.1 (Five points and a first guess)**. Given.** Five data points

$$(x_i, y_i) = (1, 3),\ (3, 7),\ (4, 9),\ (5, 16),\ (7, 20)$$

and the model $y = kx + b$. The first three points lie exactly on the line $y = 2x + 1$, so that line is a natural first guess. (The data and the lines are drawn in Example 3.1.)

Find. (a) What are $\boldsymbol{X}$, $\boldsymbol{y}$ and $\boldsymbol{k}$ for these data? (b) What is the total squared error $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e}$ of the guess $y = 2x + 1$? (c) What are $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}$, $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$ and $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{y}$, and does the boxed formula give the same error for the guess?

Answer. (a) One row per data point, with the column of ones last:

$$\boldsymbol{X} = \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \\ 7 & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 3 \\ 7 \\ 9 \\ 16 \\ 20 \end{bmatrix}, \qquad \boldsymbol{k} = \begin{bmatrix} k \\ b \end{bmatrix}.$$

There are five equations and two unknowns, and no choice of k and b satisfies all five: the first three points force $k = 2$ and $b = 1$, and that line misses $(5, 16)$.

(b) The guess has $\boldsymbol{k} = (2, 1)$ and predicts $2x_i + 1 = 3, 7, 9, 11, 15$. The errors are

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\boldsymbol{k} = (0,\ 0,\ 0,\ 5,\ 5), \qquad \boldsymbol{e}^{\mathsf{T}}\boldsymbol{e} = 0 + 0 + 0 + 25 + 25 = 50.$$

The guess fits three points exactly and misses the other two by 5 each.

(c) The entries are sums over the data, with $m = 5$:

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X} = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & m \end{bmatrix} = \begin{bmatrix} 100 & 20 \\ 20 & 5 \end{bmatrix}, \qquad \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix} = \begin{bmatrix} 280 \\ 55 \end{bmatrix}, \qquad \boldsymbol{y}^{\mathsf{T}}\boldsymbol{y} = \sum y_i^2 = 795.$$

For $\boldsymbol{k} = (2, 1)$,

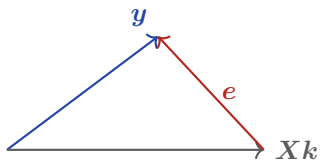
$$\boldsymbol{k}^{\mathsf{T}}(\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X})\boldsymbol{k} = 100 \cdot 4 + 2 \cdot 20 \cdot 2 \cdot 1 + 5 \cdot 1 = 485, \qquad \boldsymbol{y}^{\mathsf{T}}\boldsymbol{X}\boldsymbol{k} = 280 \cdot 2 + 55 \cdot 1 = 615,$$

so the box gives $485 - 2 \cdot 615 + 795 = 50$, as in (b). As a function of $\boldsymbol{k}$, the error depends on the data only through these three quantities. The constant $\boldsymbol{y}^{\mathsf{T}}\boldsymbol{y}$ changes the smallest value of the error, not the $\boldsymbol{k}$ where it occurs.

The next section finds the minimum from a picture.

3 A geometrical interpretation

Consider the vectors $\boldsymbol{e}$, $\boldsymbol{y}$, and $\boldsymbol{X}\boldsymbol{k}$: they all live in an m-dimensional Euclidean space $\mathbb{R}^m$ — one coordinate per *data point*, not per parameter. The total squared error $\boldsymbol{e}^{\mathsf{T}}\boldsymbol{e}$ is just the squared Euclidean distance between $\boldsymbol{y}$ and $\boldsymbol{X}\boldsymbol{k}$.



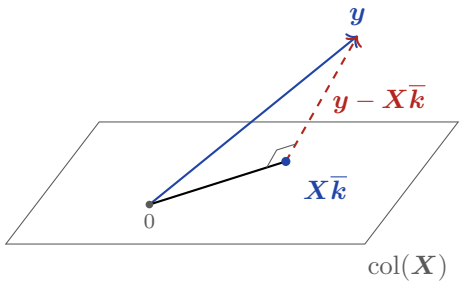
Specifically, since $\boldsymbol{X}\boldsymbol{k} \in \mathrm{col}(\boldsymbol{X})$ for every choice of $\boldsymbol{k}$, least squares fitting is finding *the point in* $\mathrm{col}(\boldsymbol{X})$ *closest to* $\boldsymbol{y}$. The column space here is the plane spanned by the two columns $(x_1, \ldots, x_m)^{\mathsf{T}}$ and $(1, \ldots, 1)^{\mathsf{T}}$ — a 2-dimensional slice of $\mathbb{R}^m$, and $\boldsymbol{y}$ almost certainly does not lie in it.

The one picture to remember

Least squares is not really about lines and data. It is: given a subspace and a point off it, drop a perpendicular. Every formula below is bookkeeping for that perpendicular.

From the diagram, the optimal $\boldsymbol{k}$, which we denote $\overline{\boldsymbol{k}}$, is such that

$$\boldsymbol{y} - \boldsymbol{X}\overline{\boldsymbol{k}} \quad \text{is orthogonal to} \quad \mathrm{col}(\boldsymbol{X}).$$



The normal equations

Orthogonal to col($\boldsymbol{X}$) means orthogonal to every column of $\boldsymbol{X}$. The columns of $\boldsymbol{X}$ are the rows of $\boldsymbol{X}^\mathsf{T}$, so the right angle in the picture reads

$$\boldsymbol{X}^\mathsf{T}(\boldsymbol{y} - \boldsymbol{X}\overline{\boldsymbol{k}}) = \boldsymbol{0},$$

that is,

$$\boxed{\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\,\overline{\boldsymbol{k}} = \boldsymbol{X}^\mathsf{T}\boldsymbol{y}} \qquad \text{(the normal equations)}.$$

They are built from the same $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ and $\boldsymbol{X}^\mathsf{T}\boldsymbol{y}$ that appeared in Section 2. When the columns of $\boldsymbol{X}$ are independent, $\boldsymbol{X}\boldsymbol{v} \neq \boldsymbol{0}$ for every $\boldsymbol{v} \neq \boldsymbol{0}$, so $\boldsymbol{v}^\mathsf{T}\boldsymbol{X}^\mathsf{T}\boldsymbol{X}\boldsymbol{v} = \|\boldsymbol{X}\boldsymbol{v}\|^2 > 0$. Then $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ is symmetric positive definite, hence invertible, and $\overline{\boldsymbol{k}} = (\boldsymbol{X}^\mathsf{T}\boldsymbol{X})^{-1}\boldsymbol{X}^\mathsf{T}\boldsymbol{y}$. For a line this fails only when all the x_i are equal.

Example 3.1 (The best line)**. Given.** The data of Example 2.1 and the sums from its part (c): $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ has rows $(100, 20)$ and $(20, 5)$, and $\boldsymbol{X}^\mathsf{T}\boldsymbol{y} = (280, 55)$.

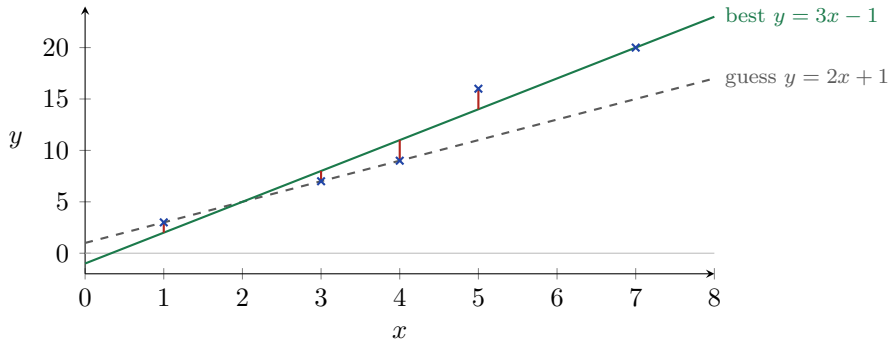
Find. (a) Which line do the normal equations give? (b) Is the residual orthogonal to col($\boldsymbol{X}$), and how does its squared length compare with the error 50 of the guess $y = 2x + 1$?

Answer. (a) The two equations are $100\,\overline{k} + 20\,\overline{b} = 280$ and $20\,\overline{k} + 5\,\overline{b} = 55$. Subtract $\frac{1}{5}$ of the first from the second: $(5 - 4)\,\overline{b} = 55 - 56$, so $\overline{b} = -1$. Then $100\,\overline{k} = 280 + 20 = 300$, so $\overline{k} = 3$. The best line is $y = 3x - 1$.

(b) The fitted values are $3x_i - 1 = 2, 8, 11, 14, 20$. The residual is the error at the best $\overline{\boldsymbol{k}}$, which we write $\overline{\boldsymbol{e}} = \boldsymbol{y} - \boldsymbol{X}\overline{\boldsymbol{k}} = (1, -1, -2, 2, 0)$. Against the two columns $(1, 3, 4, 5, 7)$ and $(1, 1, 1, 1, 1)$ of $\boldsymbol{X}$,

$$1 - 3 - 8 + 10 + 0 = 0, \qquad 1 - 1 - 2 + 2 + 0 = 0,$$

so $\overline{\boldsymbol{e}}$ is orthogonal to both columns, and hence to every combination of them, which is all of col($\boldsymbol{X}$). Its squared length is $\overline{\boldsymbol{e}}^\mathsf{T}\overline{\boldsymbol{e}} = 1 + 1 + 4 + 4 + 0 = 10$, against 50 for the guess.



The vertical segments are the entries of $\overline{\boldsymbol{e}}$ (the one at $x = 7$ is zero). The dashed guess passes through three points and still has the larger error.

4 Generalisations

Multivariate linear regression

Say our data has n independent variables,

$$\{(x_{i1}, x_{i2}, \ldots, x_{in}\,;\, y_i)\}_{i \in \{1,\ldots,m\}}.$$

Then our $\boldsymbol{X}$ matrix and $\boldsymbol{k}$ vector simply become larger:

$$\boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} & 1 \\ x_{21} & x_{22} & \cdots & x_{2n} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} & 1 \end{pmatrix}, \qquad \boldsymbol{k} = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \\ b \end{pmatrix},$$

with $\boldsymbol{X} \in \mathbb{R}^{m\times(n+1)}$ and $\boldsymbol{k} \in \mathbb{R}^{n+1}$. Nothing else changes: the same two boxed formulas hold verbatim.

Nonlinear regression

For example, what happens if I want to fit a polynomial model

$$y = \ell x^2 + mx + b$$

with parameters ℓ, m, b? Work it out in a problem set!

"Linear" means linear in the parameters

The model $y = \ell x^2 + mx + b$ is a curve, but the unknowns ℓ, m, b still enter *linearly* — so it is still a least squares problem, with the columns of $\boldsymbol{X}$ now (x_i^2), (x_i), and (1). Least squares does not care what functions of the data you put in the columns, only that the parameters multiply them linearly.

Example 4.1 (Kinetic energy of a glider on springs)**. Given.** A 0.50 kg glider on a level air track is attached to two springs, one fixed at each end, and oscillates along the track. Photogates at five positions x measure its speed as it passes, and each speed is converted to kinetic energy y:

position x (cm)	-20	-10	0	10	20
kinetic energy y (mJ)	274	673	887	861	680

With no friction, energy is conserved: $y + \frac{1}{2}\kappa\,(x - x_0)^2 = \frac{1}{2}\kappa A^2$, where κ is the combined constant of the two springs, x_0 the equilibrium position and A the amplitude. So y is a parabola in x, opening downwards.

Find. (a) What are $\boldsymbol{X}$ and the normal equations for the model $y = \ell x^2 + mx + b$? (Here m is the coefficient of x, not the number of points.) (b) Which parabola fits best? (c) What spring constant κ, equilibrium position x_0 and amplitude A does it give?

Answer. (a) The columns are x_i^2, x_i and 1, and the unknowns are (ℓ, m, b):

$$\boldsymbol{X} = \begin{bmatrix} 400 & -20 & 1 \\ 100 & -10 & 1 \\ 0 & 0 & 1 \\ 100 & 10 & 1 \\ 400 & 20 & 1 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 274 \\ 673 \\ 887 \\ 861 \\ 680 \end{bmatrix}.$$

Call the three columns of $\boldsymbol{X}$

$$\boldsymbol{c}_1 = (400, 100, 0, 100, 400), \qquad \boldsymbol{c}_2 = (-20, -10, 0, 10, 20), \qquad \boldsymbol{c}_3 = (1, 1, 1, 1, 1).$$

They are the rows of $\boldsymbol{X}^{\mathsf{T}}$:

$$\boldsymbol{X}^{\mathsf{T}} = \begin{bmatrix} 400 & 100 & 0 & 100 & 400 \\ -20 & -10 & 0 & 10 & 20 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{matrix} \leftarrow \boldsymbol{c}_1^{\mathsf{T}} \\ \leftarrow \boldsymbol{c}_2^{\mathsf{T}} \\ \leftarrow \boldsymbol{c}_3^{\mathsf{T}} \end{matrix}$$

Computing $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}$. Entry (i, j) is row i of $\boldsymbol{X}^{\mathsf{T}}$ times column j of $\boldsymbol{X}$, which is the dot product $\boldsymbol{c}_i^{\mathsf{T}}\boldsymbol{c}_j$ of two columns of $\boldsymbol{X}$:

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X} = \begin{bmatrix} \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_1 & \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_2 & \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_3 \\ \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_1 & \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_2 & \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_3 \\ \boldsymbol{c}_3^{\mathsf{T}}\boldsymbol{c}_1 & \boldsymbol{c}_3^{\mathsf{T}}\boldsymbol{c}_2 & \boldsymbol{c}_3^{\mathsf{T}}\boldsymbol{c}_3 \end{bmatrix}.$$

Since $\boldsymbol{c}_i^{\mathsf{T}}\boldsymbol{c}_j = \boldsymbol{c}_j^{\mathsf{T}}\boldsymbol{c}_i$, the matrix is symmetric and six dot products are enough. Multiply the five pairs of entries and add:

$$\begin{aligned} \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_1 &= 400^2 + 100^2 + 0^2 + 100^2 + 400^2 \\ &= 160000 + 10000 + 0 + 10000 + 160000 = 340000, \\ \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_2 &= 400(-20) + 100(-10) + 0 \cdot 0 + 100 \cdot 10 + 400 \cdot 20 \\ &= -8000 - 1000 + 0 + 1000 + 8000 = 0, \\ \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_3 &= 400 + 100 + 0 + 100 + 400 = 1000, \\ \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_2 &= (-20)^2 + (-10)^2 + 0^2 + 10^2 + 20^2 = 400 + 100 + 0 + 100 + 400 = 1000, \\ \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_3 &= -20 - 10 + 0 + 10 + 20 = 0, \\ \boldsymbol{c}_3^{\mathsf{T}}\boldsymbol{c}_3 &= 1 + 1 + 1 + 1 + 1 = 5. \end{aligned}$$

So

$$\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X} = \begin{bmatrix} 340000 & 0 & 1000 \\ 0 & 1000 & 0 \\ 1000 & 0 & 5 \end{bmatrix}.$$

The two zeros come from the symmetric positions: in $\boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{c}_2$ and $\boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{c}_3$ the terms for x and $-x$ cancel in pairs.

Computing $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$. Entry i is row i of $\boldsymbol{X}^{\mathsf{T}}$ times $\boldsymbol{y}$, the dot product $\boldsymbol{c}_i^{\mathsf{T}}\boldsymbol{y}$:

$$\begin{aligned} \boldsymbol{c}_1^{\mathsf{T}}\boldsymbol{y} &= 400 \cdot 274 + 100 \cdot 673 + 0 \cdot 887 + 100 \cdot 861 + 400 \cdot 680 \\ &= 109600 + 67300 + 0 + 86100 + 272000 = 535000, \\ \boldsymbol{c}_2^{\mathsf{T}}\boldsymbol{y} &= (-20)(274) + (-10)(673) + 0 \cdot 887 + 10 \cdot 861 + 20 \cdot 680 \\ &= -5480 - 6730 + 0 + 8610 + 13600 = 10000, \\ \boldsymbol{c}_3^{\mathsf{T}}\boldsymbol{y} &= 274 + 673 + 887 + 861 + 680 = 3375. \end{aligned}$$

The normal equations $\boldsymbol{X}^{\mathsf{T}}\boldsymbol{X}\,(\ell, m, b) = \boldsymbol{X}^{\mathsf{T}}\boldsymbol{y}$ are therefore

$$\begin{bmatrix} 340000 & 0 & 1000 \\ 0 & 1000 & 0 \\ 1000 & 0 & 5 \end{bmatrix} \begin{bmatrix} \ell \\ m \\ b \end{bmatrix} = \begin{bmatrix} 535000 \\ 10000 \\ 3375 \end{bmatrix}.$$

(b) The middle normal equation stands alone: $1000\,m = 10000$, so $m = 10$. The other two are

$$340000\,\ell + 1000\,b = 535000, \qquad 1000\,\ell + 5\,b = 3375.$$

Subtract 200 times the second from the first: $(340000 - 200000)\,\ell = 535000 - 675000$, so $140000\,\ell = -140000$ and $\ell = -1$. Then $5\,b = 3375 + 1000 = 4375$, so $b = 875$. The best parabola is

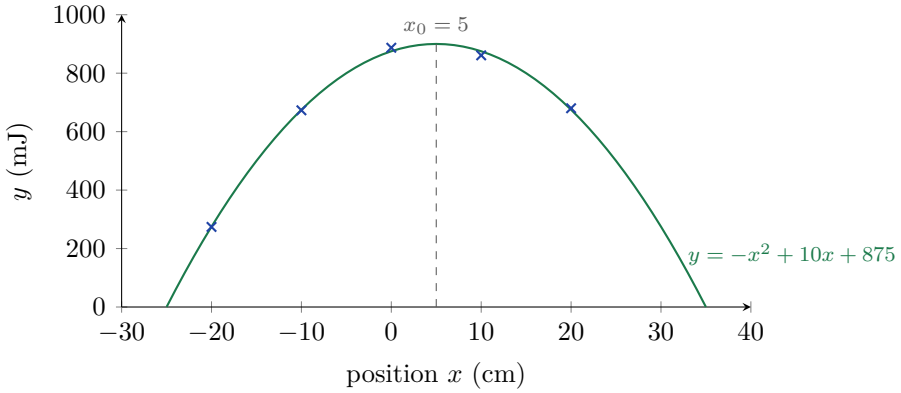
$$y = -x^2 + 10x + 875 \qquad (y \text{ in mJ},\ x \text{ in cm}),$$

and the residual is $(-1, -2, 12, -14, 5)$ mJ: no entry is larger than 14 mJ in size.

(c) Expanding the energy equation, $y = -\frac{\kappa}{2}\,x^2 + \kappa x_0\,x + \frac{\kappa}{2}\,(A^2 - x_0^2)$, so $\ell = -\kappa/2$ and $m = \kappa x_0$. Hence

$$\kappa = -2\ell = 2 \text{ mJ/cm}^2 = 20 \text{ N/m}, \qquad x_0 = \frac{m}{\kappa} = \frac{10}{2} = 5 \text{ cm},$$

using 1 mJ/cm^2 $= (10^{-3}$ J$)/(10^{-4}$ m$^2) = 10$ N/m. The largest kinetic energy, at $x_0 = 5$ cm, is $-25 + 50 + 875 = 900$ mJ $= \frac{1}{2}\kappa A^2$, which gives $A^2 = 900$ cm^2 and an amplitude $A = 30$ cm.



The three columns of $\boldsymbol{X}$ are independent because the positions take at least three different values, so $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ is invertible and the fit is unique.

Where this is going

The two ingredients we assembled today — a rectangular matrix $\boldsymbol{X}$ and the symmetric positive definite combination $\boldsymbol{X}^\mathsf{T}\boldsymbol{X}$ — reappear on Monday wearing physics. In springs and masses the rectangular matrix will be a difference matrix $\boldsymbol{A}$, a diagonal matrix $\boldsymbol{C}$ of material constants will sit in the middle, and equilibrium will read $\boldsymbol{A}^\mathsf{T}\boldsymbol{C}\boldsymbol{A}\,\boldsymbol{u} = \boldsymbol{f}$. Least squares is the case $\boldsymbol{C} = \boldsymbol{I}$.

Reading

Strang, *Computational Science and Engineering*, §2.3 (least squares for rectangular matrices).